Q5. In an Examination Hall student of second year and third year students sits in an odd and even number positions consecutively. Then a teacher asks one of the students to find the sum of even and odd positions separately. The total seating capacity of an exam hall is 100. Develop a software application to perform the above operations.

Program:

```
import java.util.*;
public class SeatingPosition {
	public static void main(String []args) {
		Scanner sc= new Scanner(System.in);
		int number, oddsum=0, evensum=0;
		int i,j;
		System.out.println("Note: The Seating Capacity of hall is 100");
		System.out.println("Enter the number of students");
		number=sc.nextInt();
		for(i=1;i<=number;i+=2)
		{
			oddsum=oddsum+1;
		}
		for(j=2;j<=number;j+=2)
		{
			evensum=evensum+1;
		}
		System.out.println("The sum of odd position is : "+oddsum);
		System.out.println("The sum of even position is : "+evensum);
	}
}
```

To compile: javac SeatingPosition.java

To execute: java SeatingPosition

Output:

Note: The Seating Capacity of hall is 100

Enter the number of students 9

The sum of odd position is : 5

The sum of even position is : 4

Q6. A Teacher asks one of the students in a class to do the arithmetic operations based on a choice with the menu driven operation. Here at a time, student have been restricted to perform one arithmetic operation only. Help the student to perform the above operations.

Requirements:

a) Display the Menu of Arithmetic operations

b) Capture the user choice on the operations

c) Capture the user inputs for the operation

d) Perform the operation based on the user choice

e) Display the result to the console.

Program:

```
import java.util.*;
public class Calculator {
    public static void main(String []args) {
        int num1=0,num2=0,option,ex;
        do
        {
            Scanner sc = new Scanner(System.in);
            System.out.println("Enter your choice from the following menu:");
            System.out.println("1.Addition 2.Subtraction 3.Multiplication 4.Division 5.Exit");
            option = sc.nextInt();
           if(option!=5){
               System.out.println("Enter the first number");
               num1=sc.nextInt();
               System.out.println("Enter the second number");
               num2=sc.nextInt();
          }
          else
               break;
           switch(option)
          {
                case 1:System.out.println("Addition of "+num1+" and "+num2+" is "+(num1+num2));
                        break;
                case 2:System.out.println("Subtraction of "+num1+" and "+num2+" is "+(num1-num2));
                        break;
                case 3:System.out.println("Multiplication of "+num1+" and "+num2+" is "+(num1*num2));
                        break;
                case 4: if(num2==0)
                                System.out.println("Error!!! In Division denominator cannot be 0!");
                        else{

       System.out.println("In division of "+num1+" by "+num2+" quotient is "+(num1/num2)+" and remainder
is "+(num1%num2));
          }
```

```
                break;
            case 5: break;
            default: System.out.println("Invalid choice"); exit(0);
        }
        System.out.println("Do you want to continue?1.Yes 2.No");
        ex=sc.nextInt();
        }while(ex==1);
}
}
```

To compile: javac Calculator.java

To execute: java Calculator

Output:

```
Enter your choice from the following menu:
1.Addition 2.Subtraction 3.Multiplication 4.Division 5.Exit
1
Enter the first number  15
Enter the second number 20
Addition of 15 and 20 is 35
Do you want to continue?1.Yes 2.No
1

Enter your choice from the following menu:
1.Addition 2.Subtraction 3.Multiplication 4.Division 5.Exit
2
Enter the first number  25
Enter the second number 10
Subtraction of 25 and 10 is 15
Do you want to continue?1.Yes 2.No
1

Enter your choice from the following menu:
1.Addition 2.Subtraction 3.Multiplication 4.Division 5.Exit
3
Enter the first number  8
Enter the second number 5
Multiplication of 8 and 5 is 40
Do you want to continue?1.Yes 2.No
1

Enter your choice from the following menu:
1.Addition 2.Subtraction 3.Multiplication 4.Division 5.Exit
4
Enter the first number  22
Enter the second number 5
```

In division of 22 by 5 quotient is 4 and remainder is 2
Do you want to continue?1.Yes 2.No
2

Q7. Input the quantity purchased and the rate. Calculate the total purchase price and display it along with the gift to be presented. The gifts to the customers are given as under:

Amount of Purchase (Rs.)	Gift
100 and above but less than 500	A key ring
500 and above but less than 1000	A leather purse
1000 and above	A pocket calculator

The application will end with a 'Thank you' message.

Program:

```
import java.util.*;
public class Gift {
	public static void main(String []args) {
		Scanner sn= new Scanner(System.in);
		int quan,price,total;
		String gift="";
		System.out.println("Enter the quantity”);
		quan= sn.nextInt();
		System.out.println("Enter the price”);
		price= sn.nextInt();
		total=price*quan;
		if(total>=100 && total<500){
			gift="a Key Ring";
		}
		else if(total>=500 && total<1000){
			gift="A leather purse";
		}
		else if(total>=1000){
			gift="a pocket calculator";
		}
		else{
			gift="not available for price range";
		}
		System.out.println(" Result :Product Price "+price+" Product Quantity "+ quan+" Total
	"+total+"rs \n Gift:"+gift);
	System.out.println("Thank you!!");
    }
}
```

To compile: javac Gift.java

To execute: java Gift

Output1:

Enter the quantity 5
Enter the price 10
Result: Product Price 10 Product Quantity 5 Total 50rs
Gift: not available for price range
Thank you!!

Output2:

Enter the quantity 15
Enter the price 20
Result: Product Price 20 Product Quantity 15 Total 300rs
Gift: A Key Ring
Thank you!!

Output3:

Enter the quantity 8
Enter the price 90
Result: Product Price 90 Product Quantity 8 Total 720rs
Gift: A leather purse
Thank you!!

Output4:

Enter the quantity 12
Enter the price 100
Result: Product Price 100 Product Quantity 12 Total 1200rs
Gift: A pocket calculator
Thank you!!

Q8. Write the following java programs using command line arguments:
Read an integer, a double, and a String from stdin, then print the values according to the instructions in the Output Format section below.

Requirement:

Input Format:

There are three lines of input:

a) The first line contains an integer.

b) The second line contains a double.

c) The third line contains a String.

Output Format:

There are three lines of output:

a) On the first line, print String: followed by the unaltered String read from stdin.

b) On the second line, print Double: followed by the unaltered double read from stdin.

c) On the third line, print Int: followed by the unaltered integer read from stdin.

Sample Input:

42

3.1415

Welcome

Sample Output:

String: Welcome

Double: 3.1415

Int: 42

Program:

```
import java.util.Scanner;
public class CommandLine {
        public static void main(String []args) {
                int i = Integer.parseInt(args[0]);
                double d =Double.parseDouble(args[1]);
                String s = args[2];
                System.out.println("String: " + s);
                System.out.println("Double: " + d);
                System.out.println("Int: " + i);
        }
}
```

To compile: javac CommandLine.java

To execute & Output:

java CommandLine 20 3456 Presidency

String: Presidency

Double: 3456.0

Int: 20

Q9. Ms. Sharada went to participate a coding contest and she has been assigned to generate the different patterns using asterisk symbol (*) one among them is to generate Diamond Shape Pattern for given number as shown below:
Sample input: if n = 4
Sample output:

```
   *
  ***
 *****
*******
 *****
  ***
   *
```

Requirement:
a) The number of rows the pattern is to be printed has to be captured
b) Using control statements generate the pattern

```
import java.io.*;
import java.util.*;

public class DiamondShape{
	public static void main(String[] args) {
		int number = 7;
		int m, n;
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter the number");
		number=sc.nextInt();
		for (m = 1; m <= number; m++) {
		// Inner loop 1 print whitespaces in between
			for (n = 1; n <= number - m; n++) {
				System.out.print(" ");
			}
		// Inner loop 2 prints star
			for (n = 1; n <= m * 2 - 1; n++) {
				System.out.print("*");
			}

		// Ending line after each row
			System.out.println();
		}
		// Outer loop 2
		 // Prints the second half diamond
		for (m = number - 1; m > 0; m--) {
		// Inner loop 1 print whitespaces inbetween
			for (n = 1; n <= number - m; n++) {
				 System.out.print(" ");
			}
		// Inner loop 2 print star
			for (n = 1; n <= m * 2 - 1; n++) {
```

```
                    System.out.print("*");
                }
            // Ending line after each row
            System.out.println();
            }
    }
}
```

To compile: javac DiamondShape.java

To execute: java DiamondShape

Output1:

```
Enter the number
4
   *
  ***
 *****
*******
 *****
  ***
   *
```